Predicting Issue Handling Process using Case Attributes and Categorical Variable Encoding Techniques

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***Abstract*—Software development process deals with the increasing complexity, needs for enhancement, and the reduction of bugs. The software project managers have a responsibility to maintain the software quality, e.g., by establishing Service Level Agreement (SLA) to the software development. However, there are difficulties to monitor SLA and to predict the software quality, due to a lot of issues should be handled. This work proposed prediction techniques for handling issues in the software development by using historical data from software repositories. There are two types of predictions in this work, namely 1) prediction of the remaining duration, and 2) prediction of the next activities. Event logs extracted from historical data in the software repositories exploited case and event attributes as predictors. Furthermore, the categorical variable encoding technique used in the preprocessing data phase. Some categorical variable encoding techniques also proposed in this study. The results showed that the use of case attributes as predictors could improve performance by 8.57% in the next activity prediction and 1.9% in the remaining duration prediction. Of the 4 (four) categorical variable encoding techniques used, one-hot encoding and sum encoding techniques can provide the best remaining duration prediction with a MAE of 19.72 and one-hot encoding technique can provide the best next activity prediction with an accuracy of 65.38%. All of these methods are widely evaluated using datasets from the Google Chromium Project. Further work including utilization of other attributes outside case and event attributes, e.g., component attributes, as predictor variables.**

Keywords— software repositories, prediction, process mining, encoding technique

I. INTRODUCTION

In its growth, software can become increasingly complex along with the need for enhancement. In addition, the appearance of bugs is also unavoidable from a software. Software maintenance could spend more than 70% of the total life cycle costs of its products [1].

In the process of handling issues, there are often operational problems related to support management systems. To measure support quality, Service Level Agreement (SLA) parameters are used. Project managers are often difficult to monitor SLAs because of the large number of issues that must be handled. In addition, many unexpected things can happen when handling an issue that causes resource utilization to be difficult. Proactive actions such as predictions can help in operational problems.

Based on these problems, a method is needed to predict the fix time of an issue. Some of the most proposed studies so far are inducing classification [2] or regression [3] from historical bug logs, encoding all available information when bugs are reported at the beginning and labeled with discrete repair time values or numerical. Therefore, all data recorded throughout the case of a bug (for example, changes made to a property such as priority, criticality, status or recipient) is completely ignored, although it can be useful for repairs, at run-time and predictions of remaining repair time.

Process mining allows automatic discovery of process models from event logs. This model provides insight and allows various types of model analysis. The process model found can be expanded with information to predict the completion time of the current event [4]. The combination of process mining techniques with data mining can improve predictive performance [5]. Polato et al. predict the remaining time using Naïve Bayes and Support Vector Regression (SVR). In Pandey's research [6], Hidden Markov Model (HMM) is used to predict the remaining time. Some business process problems from various perspectives can be solved using a combination of process mining and classification (decision tree). The combination of machine learning techniques (Artificial Neural Network), constraint satisfaction and Quality of Service (QoS) aggregation are used to improve the performance of predictions of violation findings in the ongoing process [7].

The studies mentioned above focused on the application of machine learning methods used to improve predictive performance. In [8] suggested using case attributes as predictor variables in the hope that predictive performance could be improved again. In addition, several types of categorical attributes on event logs need to be transformed into numerical forms so that data can be received by the machine learning algorithm. There is no research that specifically discusses what encoding techniques are appropriate for transforming categorical attributes into predicting the process of handling

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issues. Therefore, performance analysis is needed to determine the appropriate categorical variable encoding technique.

This paper discusses a comparative study of four categorical variable encoding techniques to be used for classification and regression using Artificial Neural Networks (ANN). There are 2 (two) types of predictions made in this paper 1) prediction of the remaining duration (regression) and 2) prediction of next activities (classification). The prediction technique for handling issues in software development using historical data from software repositories. A graphical representation of the typical application scenario of our approach is shown in Fig. 1. Specifically, given a partial trace containing the logs of already executed activities, it is possible to collect all information from case and event attributes.

The remainder of the paper is structured as follows. Section II discusses how our contribution advances the state of the art in predictive issue handling process. Section III introduces some basic concepts. Section IV defines the core framework and describes the proposed approach in detail. Implementation and experimental results, on real logs, are described in Section V, and conclusions are presented in Section VI.

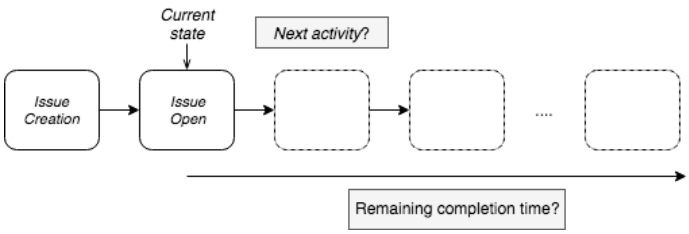

Fig. 1. Graphical representation of the application scenario of the described approach: given a running issue handling we can predict its remaining completion time and next activity.

II. Related Work

Process mining allows automatic discovery of process models from event logs. This model provides insight and allows various types of model analysis. The process model found can be expanded with information to predict the completion time of the current event [4]. In [9] presents a monitoring framework called Predictive Business Process Monitoring, namely the making of predictions and ongoing recommendations about what activities should be carried out and what data input values should be provided, so that the possibility of violating business boundaries can be minimized. For prediction, a decision tree is used to evaluate the probability for business objectives to be achieved for certain attribute value combinations. For recommendations, a decision tree is used to select a combination of attribute values that maximize the likelihood of business objectives being achieved.

In [10] conducted predictive process monitoring by approaching where traces were treated as complex symbolic sequences. The results of the research show that index-based encoding obtains higher reliability when making predictions. Therefore, index-based encoding will be used in this study.

The process mining approach has also been used to predict the timing of bug fixes by utilizing information from the issue tracking system using an annotated finite-state model [11]. However, bug fix log events are not only stored in Issue Tracking Systems (ITS), but other repositories such as Version Control Systems (VCS) and Peer Code Review Systems (PCR) also store information regarding the source code commit, patch submission and patch reviews.

Deep learning methods that are often used in natural language processing (NLP) can also be applied to business process management problems [8]. Deep learning with recurrent neural network (RNN) shows that it can improve precision predictions. However, [8] recommends determining the right way to identify cases and event attributes so that predictive performance can be improved.

III. Basic Concepts

Business processes are usually supported by an enterprise system that records data for each individual execution of a process (also called a case). Process mining is a research discipline about business process management related to obtaining insights from data execution processes (also called an event logs) that can be used to improve business processes [12].

A. Event Log

In this paper, event logs are used to form process models. Event log extraction is formed from various kinds of data sources, such as databases, message logs, transaction logs, etc. According to van der Aalst [12], the components that make up the event log are:

- The process consists of several cases.
- Case consists of several events.
- Events are in sorted cases.
- Event can have several attributes, such as activities, time, costs and resources.

B. Categorical Variable Encoding Technique

Variables on nominal scale do not have quantitative values, namely purely qualitative variables (categories or qualifications). Nominal measurement uses the level of categorical variables. Examples of nominal data are variables such as gender (male, female) and marital status (married, unmarried, divorced, separated) only showing categories.

The following are some of the techniques [13] used to encode the categorical variable:

- Ordinal encoding
  Ordinal encoding converts each value in a column to a number. It does not add any dimensions, but implies an order to the variable that may not actually exist.
- One-hot encoding
  One-hot encoding compares each level of the categorical variable to a fixed reference level. The basic strategy is to convert each category value into a new column and assigns a 1 or 0 (true/false) value to the column.

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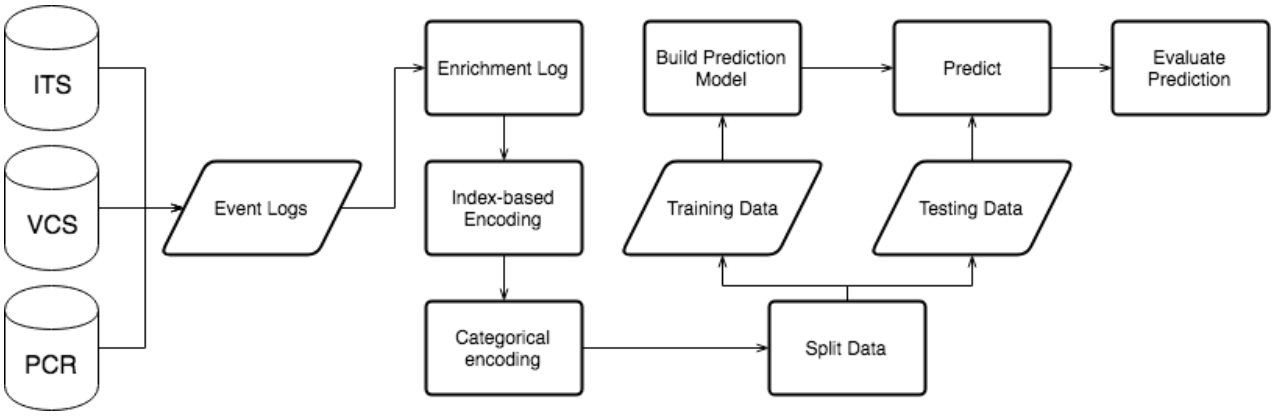

Fig. 2. Proposed method for prediction issue handling process .

- Sum encoding
  Sum encoding (also called deviation encoding) compares the mean of the dependent variable for a given level to the mean of the dependent variable for the other levels of the variable.
- Backward difference encoding
  Backward difference encoding compares the mean of the dependent variable for one level of the categorical variable to the mean of the dependent variable for the prior adjacent level.

IV. Proposed Method

In this section, the proposed approach for predictive issue handling process is described. Fig. 2 shows overview of the proposed approach. The main proposed method lies in the three stages of data processing that will be explained below.

A. Log Enrichment

Log enrichment is needed to produce output variables from the prediction analysis that will be carried out. The remaining duration variable is obtained from the calculation of the time difference between recording the current event time with the time recording at the end of the case. The next activity variable is obtained from the activity value that is executed next based on the time sequence.

B. Index-Based Encoding

In the index-based encoding, the data associated with events in a sequence is divided into static and dynamic information. Static information is the same for all the events in the sequence (e.g., the information contained in case attributes), while dynamic information changes for different events (e.g., the information contained in event attributes). The resulting feature vector g_i, for a sequence σ_i, is:

$$g_i = (s_i^1, .., s_i^u, event_{i1}, event_{i2}, ..event_{im}, h_{i1}^1, h_{i2}^1 ... h_{im}^1, ..., hr_{i1}^r, hr_{i2}^r, ... hr_{im}^r),$$

where each s_i is a static feature, each $event_{ij}$ is the event class at position j and each h_{ij} is a dynamic feature associated to an event. The example encoded data is shown in Table I.

TABLE I. Example of Index-Based Encoding

Case ID	Priority	Event_1	Event_2	…	Event_m
1	1	I_creation	0		0
1	1	I_creation	I_open		0
..	….	…	…		…

C. Categorical Variable Encoding

Part of this research dataset is categorical variable. From the original dataset there are two input variables namely type and activity. In this research, input variables will be encoded using 4 types of categorical encoding techniques, namely ordinal encoding, one-hot encoding, sum encoding and backward difference encoding. Ordinal encoding does not generate any new columns during the encoding process, but the others will increase the number of dimensions according to the number of classes in each column. In previous section, it has been explained that the activity variable will be transformed into an index-based encoding form. Therefore, the entire series of activities that have been passed will also be transformed into categorical variables with pre-determined techniques.

One of the output variables in this prediction analysis is a categorical variable. The output variable in question is the next activity. These variables are obtained through the enrichment log process that has been described in previous section. One-hot encoding technique will be used to encode output variables.

V. Implementation and Experimental Result

We conduct experiments on dataset [14] from issue tracking system (ITS), peer code review system (PCR) and version control system (VCS) of Google Chromium browser project. However, the provided dataset does not include case attributes. Therefore, we completed case attributes using crawling based on the issue id in the original dataset. Case attributes that will be used as predictors are issue type, priority and stars. Experimental dataset is shown in Table II.

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TABLE II. DATASET DESCRIPTION

Number of case	9,504
Number of event	118,853
Average number of event per case	13.77
Average completion duration per case	50.43 days
Range case creation date	30/06/2011 – 29/06/2012

For evaluating the prediction technique, we use the implementation of ANNs provided by Keras framework. Feedforward multilayer perceptron was used to build neural network for classification and regression task on the Chromium event log dataset. The algorithm was used with training for 10 epochs. The data is divided into training and testing data. Accordingly, 2/3 of data is used for training and the remaining 1/3 for testing.

TABLE III. TABLE RESULTS FOR REMAINING DURATION PREDICTION. THE RESULTS REPORTED IN MAE (DAYS). BETTER RESULT ARE HIGHLIGHTED IN BOLD

Hidden layers	Teknik categorical variable encoding	Event attributes	Case + event attributes
2	Ordinal	20,52	20,13
3		20,04	19,90
4		20,01	19,86
5		20,07	19,97
2	One-hot	19,85	**19,72**
3		20,12	20,62
4		20,54	20,71
5		21,14	21,15
2	Sum	19,85	**19,72**
3		20,42	20,57
4		20,38	20,86
5		20,33	20,37
2	Backward difference	20,03	19,92
3		20,00	19,97
4		20,28	21,80
5		20,31	20,31

The results of the prediction of the remaining duration are shown in Table III. In the table shows that the one-hot encoding technique and sum encoding with 2 hidden layers and case attributes utilization produces the best performance with MAE 19.72. It can also be seen, the use of cases and event attributes has better performance than the use of event attributes only.

The results of the prediction of the remaining duration are shown in Table IV. In the table shows that the one-hot encoding technique with 5 hidden layers and case attributes utilization produces the best performance with an accuracy 65.38%. It can also be seen, the use of cases and event attributes has better performance than the use of event attributes.

TABLE IV. TABLE RESULTS FOR NEXT ACTIVITY PREDICTION. THE RESULTS REPORTED IN ACCURACY (%). BETTER RESULT ARE HIGHLIGHTED IN BOLD

Hidden layers	Teknik categorical variable encoding	Event attributes	Case + event attributes
2	Ordinal	60,39	60,27
3		62,24	62,46
4		62,02	61,35
5		60,92	62,65
2	One-hot	62,96	63,75
3		64,98	64,84
4		64,53	64,09
5		64,22	**65,38**
2	Sum	64,73	64,88
3		61,17	64,44
4		57,67	64,73
5		63,79	64,64
2	Backward difference	62,95	62,90
3		54,60	63,17
4		61,09	62,88
5		62,78	62,39

VI. CONCLUSION

In this paper we proposed a new approach for the prediction of issue handling process. The prediction exploited case and event attributes. The extracted event log is processed using three preprocessing stages: 1) enrichment log, 2) index-based encoding and 3) categorical variable encoding.

The results show that case attributes of the event log can affect predictive performance results. The use of data containing case attributes on average can result in better predictive performance. In the case attributes utilization in the next activity prediction can improve the predictive performance by 8.57% and in the remaining duration prediction can increase by 1.9%.

Categorical variable encoding techniques can affect the results of predictive performance. One-hot encoding and sum encoding technique can produce the best performance compared to other techniques. In the remaining duration prediction, one-hot encoding and sum encoding can produce a MAE value of 19.72. In the next activity prediction, one-hot encoding can produce an accuracy value of 65.38%.

For further development, the use of other data sources can be done with the aim of performance comparison. Things that need to be considered in the use of different data sources are in the process of collecting data. In addition, there needs to be further research related to the utilization of other attributes both case and event attributes. Case attributes such as component or event attributes such as resources should be considered to be used as predictor variables.

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